

Zhenyang Huang

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University of Chinese Academy of Sciences, Xinjiang Astronomical Observatory

EDUCATION

- **Max-Planck-Institut für Kernphysik (MPIK), Heidelberg University** Oct.2026 - present
Ph.D. Student in Astrophysics
- **University of Chinese Academy of Sciences** Aug.2023 - Jun.2026
M.S. in Astronomy Xinjiang Astronomical Observatory
- **Sun Yat-sen University** Aug.2016 - Jun.2020
B.S. in Physics School of Physics and Astronomy

PUBLICATIONS

* = CO-FIRST AUTHOR

- [1] Zhenyang Huang, Haihao Shi, Zhiyong Liu, Na Wang. (2025). **An Interpretable AI Framework to Disentangle Self-Interacting and Cold Dark Matter in Galaxy Clusters: The CKAN Approach.** *The Astronomical Journal (AJ)*, 170, 263 . [doi:10.3847/1538-3881/ae0476](https://doi.org/10.3847/1538-3881/ae0476).
- [2] Haihao Shi*, Zhenyang Huang*, Qiyu Yan, Jun Li, Guoliang Lü, Xuefei Chen. (2025). **Hunting Hidden Axion Signals in Pulsar Dispersion Measurements with Machine Learning.** *The Astrophysical Journal (ApJ)*, 995, 82 . [doi:10.3847/1538-4357/ae18d1](https://doi.org/10.3847/1538-4357/ae18d1).
- [3] Haihao Shi*, Zhenyang Huang*, Qiyu Yan, Junda Zhou, Guoliang Lü, Xuefei Chen. (2026). **Application of interpretable data-driven methods for the reconstruction of supernova neutrino energy spectra following fast neutrino flavor conversions.** *Physics Letters B (PLB)*, 875, 140371 . [doi:10.1016/j.physletb.2026.140371](https://doi.org/10.1016/j.physletb.2026.140371).
- [4] Haihao Shi, Zhenyang Huang, Junda Zhou, Guoliang Lü, Xuefei Chen. (2025). **A Core-Collapse Supernova Neutrino Parameterization with Enhanced Physical Interpretability.** *The Astrophysical Journal Supplement Series (ApJS)*, 282, 48. [doi:10.3847/1538-4365/ae231f](https://doi.org/10.3847/1538-4365/ae231f).
- [5] Junda Zhou, Zhenyang Huang, Di-Chang Chen, Jianheng Guo. (2026). **Inversion of Hydrogen-rich Atmosphere and Water Content for GJ 486b.** *The Astrophysical Journal (ApJ)*, 1002, 151 . [doi:10.3847/1538-4357/ae5e6c](https://doi.org/10.3847/1538-4357/ae5e6c).
- [6] Haihao Shi, Junda Zhou, Zhenyang Huang, Guoliang Lü, Xuefei Chen. (2025). **Dark Matter (S)pins the Planet.** *Journal of Cosmology and Astroparticle Physics (JCAP)*. [doi:10.1088/1475-7516/2026/01/035](https://doi.org/10.1088/1475-7516/2026/01/035).

CONFERENCES & PRESENTATIONS

- **Chinese Astronomical Society Annual Meeting (CAS 2025)** Nov. 2025
Xiamen, China
 - **Oral Presentation:** *An Interpretable AI Framework to Disentangle Self-Interacting and Cold Dark Matter in Galaxy Clusters: The CKAN Approach* [[The Astronomical Journal](https://doi.org/10.3847/1538-3881/ae0476), 170, 263 (2025)]
- **11th Youth Astronomical Forum of the Chinese Astronomical Society (YAF-11)** Aug. 2025
Dali, China
 - **Poster Presentation:** *Application of Interpretable Data-Driven Methods for the Reconstruction of Supernova Neutrino Energy Spectra Following Fast Neutrino Flavor Conversions* [[arXiv:2507.09632](https://arxiv.org/abs/2507.09632)]

PROJECTS

- **Gravitational Wave Data Exploration: A Practical Training in Programming and Analysis** Nov.2023 - Jan.2024
ICTP, UCAS
 - Developed machine learning model for gravitational wave data exploration
 - Led the team to win the championship of the final Kaggle competition of the project

SKILLS

- **Programming Languages:** Python (primary), C, C++, CUDA C/C++
- **Data Science & Machine Learning:** PyTorch, NumPy, Astropy, Scikit-learn, TensorFlow
- **Other Tools & Technologies:** Docker, Linux (Ubuntu), Shell Scripting, LaTeX
- **Languages:** English (conversational), Cantonese Chinese (native), Mandarin Chinese (native)